

Physics

1 → ~~Conver~~ lens is used to produce real image.

2 →

Convex lens of focal length 10 cm has greater power with power +10D.

⇒

3 → solution —

$$u = -5 \text{ cm}$$

$$v = -10 \text{ cm}$$

$$f = ?$$

$$\Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \Rightarrow \frac{1}{10} - \left(\frac{-1}{5}\right) = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1}{10}$$

$$\Rightarrow f = 10 \text{ cm}$$

Ans → It is a convex lens with focal length 10 cm.

ID

Shriving Yadav

10 = 11

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Ans → It is a converge lens with focal length 10 cm.

$$4 \rightarrow v = 50 \text{ cm}$$

$$\text{Height of image} = h_i$$

$$\text{" " object} = h_o$$

$$h_i = h_o$$

$$\text{Magnification, } m = -1$$

$$u = ?$$

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Height of image = h_i

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$$h_i = h_o$$

Magnification, $m = -1$

$$u = ?$$



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Chemistry —

$$m = \frac{v}{u}$$

$$\Rightarrow -1 = \frac{50}{u}$$

$$\Rightarrow u = -50 \text{ cm}$$

$$f = ?$$

By Lens Formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{50} + \frac{1}{50} = \frac{2}{f}$$

$$f = \frac{50}{2} = 25 \text{ cm}$$

$$\text{Power of lens} = \frac{1}{f}$$

$$= \frac{100}{25}$$

$$= +4D$$

Ans → The object distance from convex lens is 50 cm
f power of lens is 4D.

Chemistry,

→ HCl — Hydrochloric acid

H₂SO₄ → Sulphuric acid

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Ans $\rightarrow$ The object distance from convex lens is 50cm
& power of lens is 4D.

Chemistry.

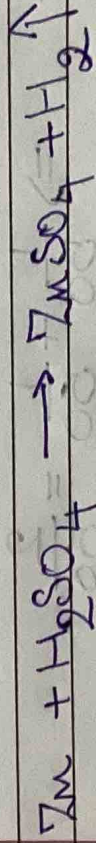
1 → HCl - Hydrochloric Acid
 H_2SO_4 → Sulphuric Acid

2 → Turmeric is an natural acid, base indicator.
On adding to basic solution colour change to reddish brown. On adding it to acidic solution colour changes to yellow.

Exp

3 → (a) Acid + Metal $\Rightarrow$ Salt + ~~Hydrogen~~ Hydrogen

When zinc granules are added to sulphuric acid



→ Zinc sulphate is formed & hydrogen gas is released.

b) Hydrogen gas. It could be tested by bringing burning candle near it and will sound pop.

Biology
1 → Left Atrium

2 → Aorta

2 → Turmeric is a natural acid, base indicator.
On adding to basic solution, colour changes to reddish brown. On adding it to acidic solution colour changes to yellow.



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4 → Three functions of lymph are

- Protecting body from illness causing invaders.
- Maintaining body fluid levels.
- Absorbing digestive tract fat & removing cellular waste.

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